SPE-229159-MS

Accelerating CCUs Storage Site Screening Using AI/ML: A Data-Driven Approach to Early-Stage Subsurface Evaluation

S. Gupta, Rara Energy Consulting, Abu Dhabi, United Arab Emirates; A. Suboyin, ADNOC Logistics & Services, Abu Dhabi, United Arab Emirates; D. Ranjan, Rara Energy Consulting, Abu Dhabi, United Arab Emirates; A. Chand, Indian Institute of Technology, Indian School of Mines, Dhanbad, India; O. Baobaid, ADNOC Logistics & Services, Abu Dhabi, United Arab Emirates

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

As the energy landscape shifts toward sustainability, the role of subsurface gas storage particularly in porous geological reservoirs has become a necessity for carrying the future of energy systems such as hydrogen, CO_2, and natural gas storage. However, the selection of suitable underground sites, especially depleted fields and aquifers, remains demanding and challenging. Traditional approaches rely on manual data mining, quality control, analytics, and prolonged report reviews, placing significant demands on subsurface specialists, inflating costs, and delaying project timelines. Addressing these limitations, this paper introduces an AI and machine learning framework for streamlining early-stage screening of gas storage sites. The methodology was validated across more than ten gas fields, with an AI/ML-assisted comparative analysis leading to focused evaluations specifically, Chance of Success (CoS) calculations on the most promising candidates. The approach demonstrates its value through assessments of sites spanning a wide risk spectrum, with cases such as structurally robust anticlines facing minimal aquifer risk and, conversely, sites prone to aquifer interactions and complex containment challenges.

AI-driven models in this workflow deliver enhanced subsurface risk assessment by detecting faults, seal weaknesses, fluid anomalies, and aquifer signatures using large historical datasets. This enables the elimination of zones with unmitigable risks and drive site selection with greater confidence. The integration of structural integrity, reservoir continuity, aquifer proximity, and analog geodata further strengthens the models ability to calculate CoS scores with embedded uncertainty, streamlining the ranking process for a diverse set of geological storage candidates.

Preliminary project experience demonstrates that this approach can reduce the timeline of the initial site screening phase by 20-30%. This helps geoscience and reservoir engineering teams to focus their efforts and resources on locations with the highest potential, while complementing established simulation workflows, rather than replacing them. The proposed AI framework thus offers a rapid, scalable, and repeatable screening process, automating risk checks and blending advanced analytics with domain expertise. Ultimately, this study advances in subsurface site selection by accelerating business decisions and

supporting the global shift to cleaner energy through smarter, faster geological screening and de-risking, directly fueling progress toward net-zero targets.

Introduction

As global efforts to achieve net-zero emissions intensify, Carbon Capture, Utilization, and Storage (CCUS) has emerged as an indispensable technology for decarbonizing hard-to-abate industries. The success of the entire CCUS value chain, however, hinges on a critical, often-overlooked bottleneck: the rapid and reliable identification of suitable geological storage sites. The ability to securely and permanently sequester vast quantities of CO_2 depends entirely on the quality of the subsurface reservoirs chosen, making the initial screening phase a high-stakes decision point that dictates the technical viability, economic feasibility, and long-term safety of any project.

Screening potential fields for CO_2 storage is one of the most challenging and resource-intensive stages in a CCUS feasibility study. It requires integrating large volumes of geological, geophysical, and operational data while applying stringent selection criteria to ensure that only technically suitable and economically viable sites move forward. This process involves evaluating subsurface integrity, storage capacity, injectivity, and long-term containment risks, all under conditions of uncertainty and limited available data.

As shown in Table 1, the preliminary site screening alone can take 1–2 months, representing nearly 30% of the total project timeline and cost in a typical 8-month CCUS storage feasibility study. The challenge is no longer just finding a site, but efficiently screening hundreds of potential candidates across vast regions to build a global portfolio of secure storage assets.

Table 1—Timeline for a typical CCUs subsurface evaluation study *(Internal Document, Rara Energy Consulting 2024)*

Phase	Task	Description	Months							
			1	2	3	4	5	6	7	8
I	1	Framing session: Definitions of work scope and deliverables. Subsurface Data overview								
I	2	Assembling and integrating data. Conduct Field/Well performance analysis								
I	3	Preliminary Regional Site screening								
II	3	G&G and Simulation Work. History matching Work and injection Scenarios (Based on existing models) or building sector models.								
II	5	Risk and Uncertainty analysis. Model results optimisation								
III	6	Conceptual Development Plan								
III	7	Economic analysis								
IV	8	*Report Writing and Project Close-out								

The paradigm is shifting with the convergence of two powerful forces: the increasing availability of large-scale, digitized geoscience datasets and the maturation of artificial intelligence (AI) and machine learning (ML). These technologies offer the potential to transform site screening from a qualitative art into a quantitative, data-driven science, enabling faster, more consistent, and more comprehensive evaluations.

By training algorithms to recognize the complex, nonlinear relationships between geological attributes and storage suitability, we can augment human expertise and dramatically accelerate the screening workflow.

This paper introduces a novel framework that leverages this synergy. We present an AI/ML methodology designed to accelerate the early-stage evaluation of CO_2 storage sites by predicting the Geological Chance of Success. Using a comprehensive public dataset from 65 sedimentary basins compiled by the Australian Petroleum Cooperative Research Centre's GEODISC program, we demonstrate how machine learning can systematically analyze key geological parameters from porosity and permeability to seal integrity and trap configuration. To validate this approach, we developed and benchmarked a suite of machine learning models, including deep neural networks, tree-based ensembles (Random Forest, XGBoost), and traditional regression techniques. The following sections detail our methodology, present a comparative performance analysis, and discuss the implications of our findings for accelerating the global deployment of CCUS.

Methodology

Approach. The approach is structured into four key stages: data acquisition and curation, feature engineering and selection, model development and training, and performance evaluation and validation. The data was sourced from the GEODISC program under the Australian Petroleum Cooperative Research Centre, covering 65 sedimentary basins and capturing essential subsurface and geospatial attributes. Following data quality assessment, feature engineering was performed using Pearson's correlation coefficient to isolate features with the highest predictive power for geological success. A threshold of $|r| \geq 0.3$ was applied to retain features with meaningful linear correlation, while ensuring only geologically relevant variables were considered.

To ensure model generalizability and prevent overfitting, a rigorous 5-fold cross-validation technique was employed on the training dataset, which was split using an 85%-15% train-test ratio. Within each fold, stratified sampling and feature scaling were applied to maintain data balance and consistency. A broad spectrum of algorithms was explored, including Deep Neural Networks[1], Multi-Layer Extreme Learning Machines (ML-ELM)[2], Random Forest[3], XGBoost[4], and Linear Regression. Multiple hyperparameter configurations were tested for each neural network, varying the number of hidden layers, neurons, activation functions, and regularization techniques. Performance was assessed using industry-standard metrics such as R^2, MAE, MSE, and RMSE. Python was the primary programming language due to its extensive ecosystem of data science libraries and community support. Data preprocessing, statistical analysis, and feature engineering were conducted using Pandas, NumPy, and SciPy. Scikit-learn provided essential utilities for data splitting, feature scaling, correlation analysis, and evaluation metrics. For model development, TensorFlow and Keras were used to build and train deep neural networks, offering flexibility in configuring various architecture. The Multi-Layer Extreme Learning Machines (MELM) models were implemented using custom- made class. Linear Regression and Tree-Based algorithms, including Random Forest and XGBoost, were developed using their respective implementations in Scikit-learn and XGBoost libraries. All training and evaluation experiments were executed on cloud-based GPU infrastructure powered by NVIDIA T4 Tensor Core GPUs.

Data Federation. Datasets from the United Kingdom and the United States were initially explored for use in this study. However, they were found unsuitable for the machine learning workflow. A major constraint was the absence of a clearly defined and quantifiable target variable required for supervised learning.

The dataset used for training machine learning models was collected from sites identified across all Australian sedimentary basins. The availability of a clearly defined target variable—*Geological Chance of Success*—enhanced its suitability for supervised machine learning applications. These sites were potential locations for geological sequestration, and the data were gathered and compiled under the Australian

Petroleum Cooperative Research Centre's GEODISC program[10]. The dataset is publicly accessible on the Geoscience Australia Portal. The study was conducted on 65 sedimentary basins.

Fig. 1 and Fig. 2 contains depth wise and ESSCI (Environmentally Sustainable Sites for Carbon Dioxide Injection) type-wise plotting of all 65 fields on the Australian Map, respectively. On performing data quality analysis, features having null values were identified and 2 sites had missing data for Average Porosity, and were removed from the sample.

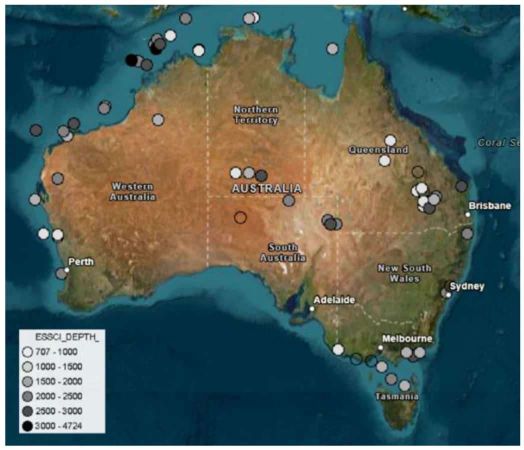

Figure 1—65 Sites: Depth-wise labelling

Figure 2—65 Sites shown Type-wise labelling

Feature Engineering

Feature selection is a crucial step in machine learning that can significantly impact model performance. As the number of features increases, the volume of the feature space grows exponentially. This means your data becomes increasingly sparse, making it harder for algorithms to find meaningful patterns (Curse of Dimensionality). Also, irrelevant or redundant features can cause models to learn noise rather than signal, leading to poor generalization(Overfitting Prevention)[5].

Pearson's correlation coefficient, often symbolized as r, is a fundamental statistical measure that quantifies the strength and direction of a linear relationship between two continuous variables. The equation used is shown as Eq. 1. The absolute value of r indicates the strength of the linear relationship. $|r|$ close to 1 would indicate a strong linear relationship. $|r|$ close to 0 would indicate a weak or no linear relationship[6]. So it was used for selecting the important features.

The value of Pearson's correlation coefficient of all of the numerical features with respect to geological chance is shown in Fig. 3.

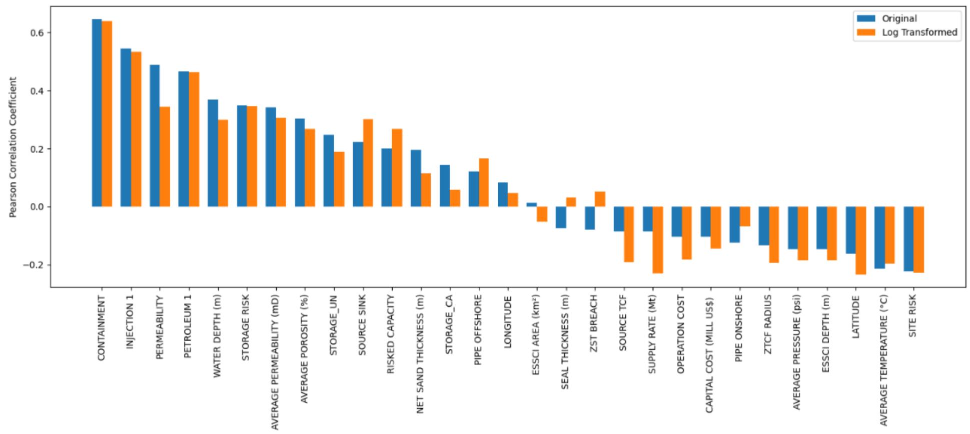

Figure 3—Pearson's Correlation Coefficient with Geological Chance

A threshold value |r|>=0.3 was considered for selection. 8 following features were selected as an input f or model development:

- CONTAINMENT
- INJECTION 1
- PERMEABILITY
- PETROLEUM 1
- WATER DEPTH (m)
- STORAGE RISK
- AVERAGE POROSITY (%)
- log(SOURCE SINK)

'SOURCE SINK', as per definition, is not logically related to the Geological Chance of Success. Since only geological factors should be considered, not proximity to a source or monetary factors, therefore, 'SOURCE SINK' was excluded, and consequently 7 features were finalized for model's training. Fig. 4 represent Pearson's coefficient value of these 7 features.

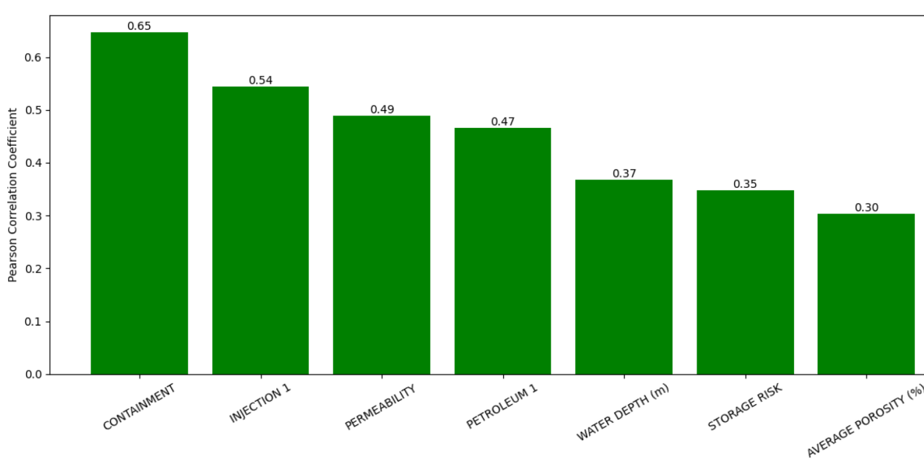

Figure 4—Final features selected with their Pearson's correlation coefficient

Detailed definition of every feature utilised in the study is given in Table 2. A comprehensive statistical overview of the selected features employed in this research is presented in Table 3 and Fig. 5.

Table 2—Detailed definition of features in the dataset.

Feature	Definition
Basin	A low area in the Earth's crust, of tectonic origin, in which sediments accumulate.
Play	An area in which hydrocarbon accumulations occur
ESSCI Type	The trapping mechanism of the ESSCI area
ESSCI Depth	Depth of the ESSCI
Average Porosity	Average porosity of the ESSCI area in %
Average Permeability	Average permeability of the ESSCI area in millidarcies (mD)
Net Sand Thickness	Thickness of reservoir quality lithology
ESSCI Area	Size of the ESSCI in square kilometres (km^2)
Seal Lithology	Type of lithological unit overlying the reservoir
Seal Thickness	Thickness of the lithological unit overlying the reservoir in metres
Geological Chance	Each basin was ranked on the following risk factors to determine geological chance: storage capacity, injectivity potential, containment and existing natural resources
Trap Type 1	Type of lithological or structural trap overlying the reservoir prohibiting vertical or lateral migration of reservoir contents
Trap Type 2	Second type of lithological or structural trap overlying the reservoir prohibiting vertical or lateral migration of reservoir contents
Average Temperature	Average Temperature of the entire ESSCI in degrees Celsius (°C)
Average Pressure	Average Pressure of the entire ESSCI in millidarcies (mD)
Injection	Injectivity Potential: The chance that reservoir conditions will be viable for injection
CO2 Source	The source location of CO2
Source TCF	Amount of CO2 from the source area in trillion cubic feet
Supply Rate	Approximate amount of CO2 in million tonnes that could be supplied to an ESSCI site over 20 years
Permeability	Measurement of a rock's ability to transmit fluids, measured in millidarcies (mD)
Injection 1	The risk assigned for there being suitable reservoir conditions for CO2 injection. The smaller the number, the greater the risk that reservoir conditions are unsuitable
Storage_UN	Storage Parameter
Storage Capacity	The total pore volume within the ESSCI
Storage Risk	The storage risk is the overall chance multiplied by the estimated storage capacity
Site Risk	The chance that the site is economically and technically viable
ZTCF Radius	The radius that a 1 Trillion Cubic Feet (TCF) injection of CO2 would take up under reservoir temperature and pressure conditions
ZST Breach	Distance in kilometre (km) to the nearest potential breach (fault or edge of seal)
Containment	The chance that the seal and trap will be effective for sequestering CO2
Petroleum	The likelihood that hydrocarbons are present in the ESSCI area
Petroleum 1	The risk that was assigned for the presence of existing hydrocarbon resources. The lower the number the greater the risk that there would be existing resources impacted by CO2 storage
Overall Chance	Product of all five individual ESSCI factors (Storage, Injection, Site, Resources, and Containment)
Risked Capacity	ESSCI chance multiplied by the total estimated storage capacity of CO2
Source Sink	Distance in kilometre (km) from the nearest major CO2 source to the nearest viable ESSCI
Pipe Offshore	Length of pipeline in kilometre (km) required from the identified near major source to the offshore storage site

Feature	Definition
Pipe Onshore	Length of pipeline in kilometre (km) required from the identified near major source to the onshore storage site
Capital Cost	Initial cost in million US dollars

Table 3—Statistical Overview of the Dataset

Feature Name	Mean	Standard Deviation	Minimum	Maximum
WATER DEPTH (m)	75.07	142.62	0.00	851
AVERAGE POROSITY (%)	16.81	5.68	3.60	30
PERMEABILITY	65.84	116.41	0.02	750
INJECTION 1	0.74	0.21	0.10	1
STORAGE RISK	0.87	0.19	0.20	1
CONTAINMENT	0.61	0.28	0.10	1
PETROLEUM 1	0.71	0.23	0.10	1
Target Variable				
GEOLOGICAL CHANCE	0.29	0.22	0.02	1

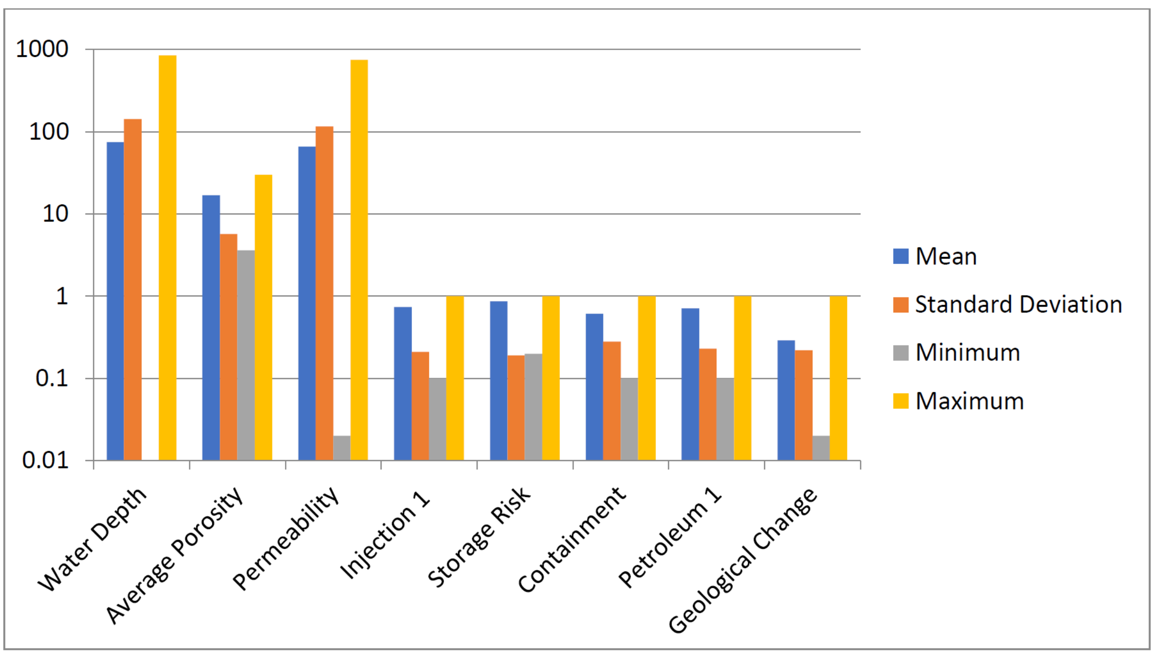

Figure 5—Statistical Overview of the Dataset

Model Sampling and Training. The dataset was split 85%-15% for training/cross-validation and testing respectively. **K-fold cross-validation** was employed as a robust resampling procedure to reliably evaluate model performance and avoid bias from single train-test splits.

The training set was partitioned into K=5 equally-sized, non-overlapping folds as shown in Fig. 6. This process was repeated 5 times. Over each iteration, each fold serves as validation data while the remaining 4 folds form the training set, with performance scores averaged for a comprehensive estimate. This ensures every data point contributes to both training and testing, providing more stable and less variable performance metrics through averaging across iterations. K-fold cross-validation maximizes dataset utilization, which is particularly advantageous for smaller datasets where large hold-out sets would compromise training.

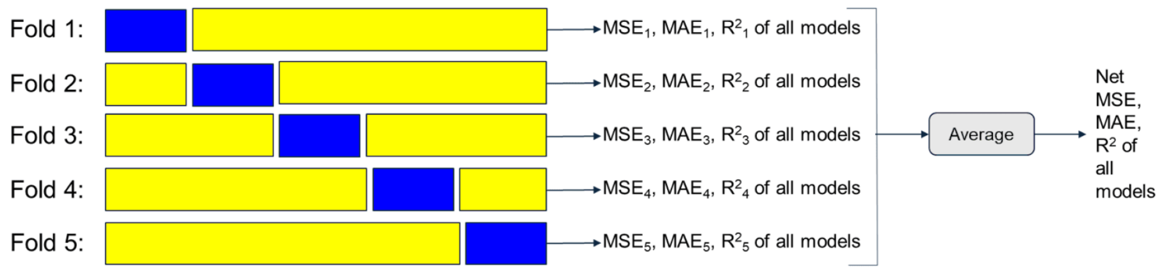

Figure 6—5-fold cross-validation technique

Within each fold, training data was expanded through repetition to help models, especially neural networks, learn more effectively from limited samples. Features were scaled to ensure comparable scales, with scalars fitted only on training data to prevent data leakage.

Multiple configurations of **Deep Neural Networks, Multi-Layer Extreme Learning Machines (MELM)**, **Linear Regression, Random Forest**, and **XGBoost** were trained and evaluated using this framework.

Performance Metrics. Various error metrics are used to quantify the difference between the predicted values and the actual observed values.

- MAE (Eq. 1) is the average of the absolute differences between the predicted and actual values. It measures the average magnitude of the errors in a set of predictions, without considering their direction.
- MSE (Eq. 2) is the average of the squared differences between the predicted and actual values. Squaring the errors means that larger errors are penalized much more heavily than smaller ones. This makes MSE sensitive to outliers.
- RMSE (Eq. 3) is the square root of the MSE. It essentially brings the error metric back to the same units as the target variable, making it more interpretable than MSE.
- R^2 (Eq. 4) measures the proportion of variance explained by the model, so higher values (closer to 1.0) indicate better performance.

$$MAE = \frac{1}{n}\sum_{i=1}^{n} |y_i - \widehat{y_i}|$$

Eq. 1—Mean Average Error's Formula

$$\mathrm{MSE} = \frac{1}{n}\sum (Y_i - \hat{Y}_i)^2$$

Eq. 2—Mean Squared Error's Formula

$$RMSE = \sqrt{\sum_{i=1}^{n} \frac{(\hat{y}_i - y_i)^2}{n}}$$

Eq. 3—Root Mean Squared Error's Formula

$$R^2 = 1 - \frac{\sum_i (y_i - \hat{y}_i)^2}{\sum_i (y_i - \overline{y})^2}$$

Eq. 4—R-Squared Error's Formula

Results and Discussion

Model Performance on Training Data. The performance metrics for various combinations of algorithms and hyperparameters fine tuning, averaged across the 5 folds, is shown in Table 4 and Fig.7.

Table 4—Average performance of the models on the training data across all 5 folds.

Model	R^2	MSE	MAE	RMSE
Neural_Network_Config_1	0.92	0.0037	0.047	0.060
Neural_Network _Config_2	0.98	0.0011	0.025	0.032
Neural_Network _Config_3	0.93	0.0034	0.045	0.058
Neural_Network _Config_4	0.97	0.0013	0.027	0.036
Neural_Network _Config_5	0.97	0.0012	0.026	0.036
Neural_Network _Config_6	0.92	0.0037	0.047	0.060
Neural_Network _Config_7	0.97	0.0015	0.031	0.039
Neural_Network _Config_8	0.97	0.0015	0.029	0.039
Neural_Network _Config_9	0.57	0.0205	0.094	0.115
Neural_Network _Config_10	0.96	0.0019	0.033	0.043
Neural_Network _Config_11	0.96	0.0018	0.033	0.043
Linear Regression	0.91	0.0044	0.052	0.066
ML_ELM_Config_1	0.95	0.0026	0.039	0.050
ML_ELM_Config_2	0.78	0.0103	0.077	0.100
ML_ELM_Config_3	0.71	0.0137	0.086	0.116
Random Forest	1.00	0.0000	0.000	0.000
XGBoost	1.00	0.0000	0.000	0.000

Tree-based algorithms like Random Forest and XGBoost have R^2 scores much higher than Neural Networks in the training set (often approaching 1), indicating potential overfitting where these models memorize training data patterns rather than learning generalizable features, which typically results in poorer performance on unseen test data. Most neural network configurations show strong but not perfect performance (R^2 ranging from 0.91146 to 0.97498), suggesting they are learning meaningful patterns without complete overfitting. ML-ELM configurations exhibit lower R^2 scores (0.77611 to 0.92324), with ML-ELM_Config_1 performing best (R^2 = 0.92324). This suggests the random weight initialization approach may be limiting optimal feature learning. Standard Linear Regression shows reasonable performance (R^2 = 0.90608), providing a good baseline for comparison.

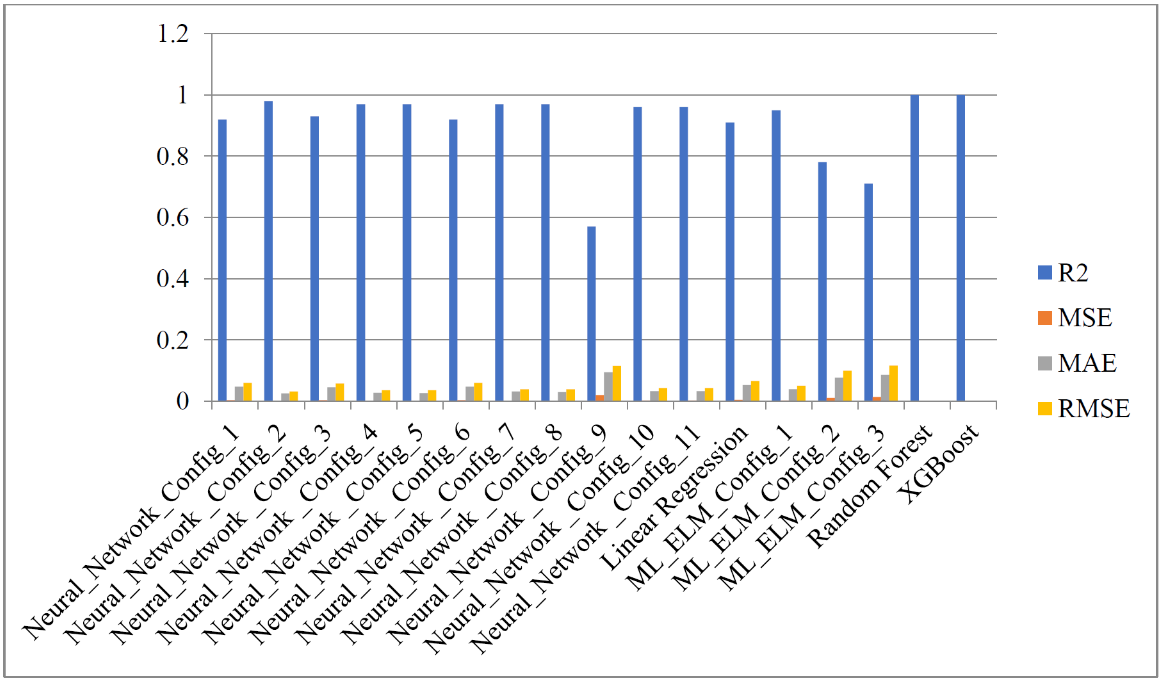

Figure 7—Average performance on the training data across all 5 folds as a bar chart[9].

Model Performance on Validation Data. The performance metrics for all five models on the validation data, averaged across the 5 folds, is shown in Table 5 and Fig. 8. Most neural network configurations maintain R^2 scores above 0.90, indicating strong generalization capability.

Table 5—Model's average performance on the validation data across all 5 folds[9]

Model	R^2	MSE	MAE	RMSE
Neural_Network_Config_1	0.88	0.0053	0.057	0.071
Neural_Network _Config_2	0.95	0.0024	0.036	0.045
Neural_Network_Config_3	0.88	0.0052	0.059	0.070
Neural_Network_Config_4	0.94	0.0026	0.037	0.048
Neural_Network_Config_5	0.94	0.0027	0.038	0.048
Neural_Network_Config_6	0.88	0.0052	0.054	0.070
Neural_Network_Config_7	0.93	0.0032	0.042	0.052
Neural_Network_Config_8	0.94	0.0028	0.039	0.050
Neural_Network_Config_9	0.57	0.0186	0.087	0.110
Neural_Network_Config_10	0.93	0.0033	0.043	0.053
Neural_Network_Config_11	0.93	0.0031	0.040	0.051
Linear Regression	0.84	0.0072	0.065	0.066
ML_ELM_Config_1	0.12	0.0432	0.092	0.081
ML_ELM_Config_2	0.59	0.0197	0.107	0.162
ML_ELM_Config_3	0.38	0.0304	0.127	0.136
Random Forest	0.65	0.0167	0.098	0.126
XGBoost	0.48	0.0229	0.111	0.150

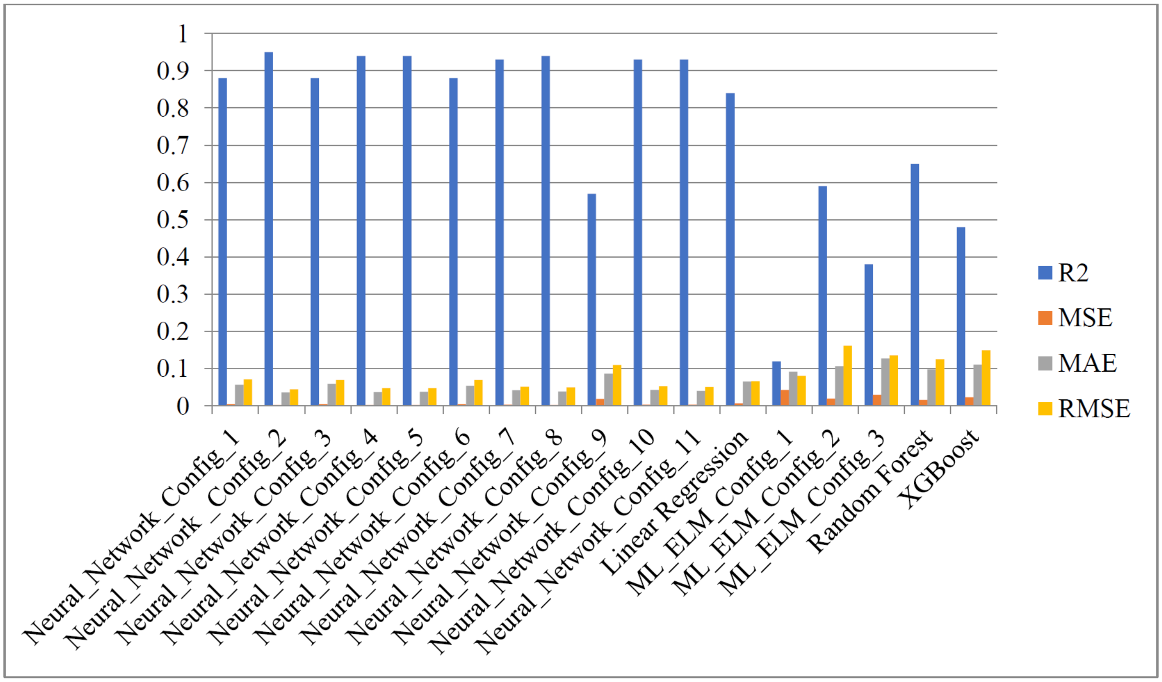

Figure 8—Model's average performance on the validation data across all 5 folds[9]

Comparative Analysis of Generalization and Overfitting. The variation in training and validation R^2 and RMSE values is demonstrated in Fig. 9 and Fig.10, respectively. These visualizations reveal that superior training performance does not necessarily correlate with effective generalization capability. The magnitude of the disparity between training and validation R^2 scores serves as an indicator of model generalization: a minimal gap signifies robust generalization performance, whereas a substantial gap indicates overfitting and compromised model reliability on unseen data.

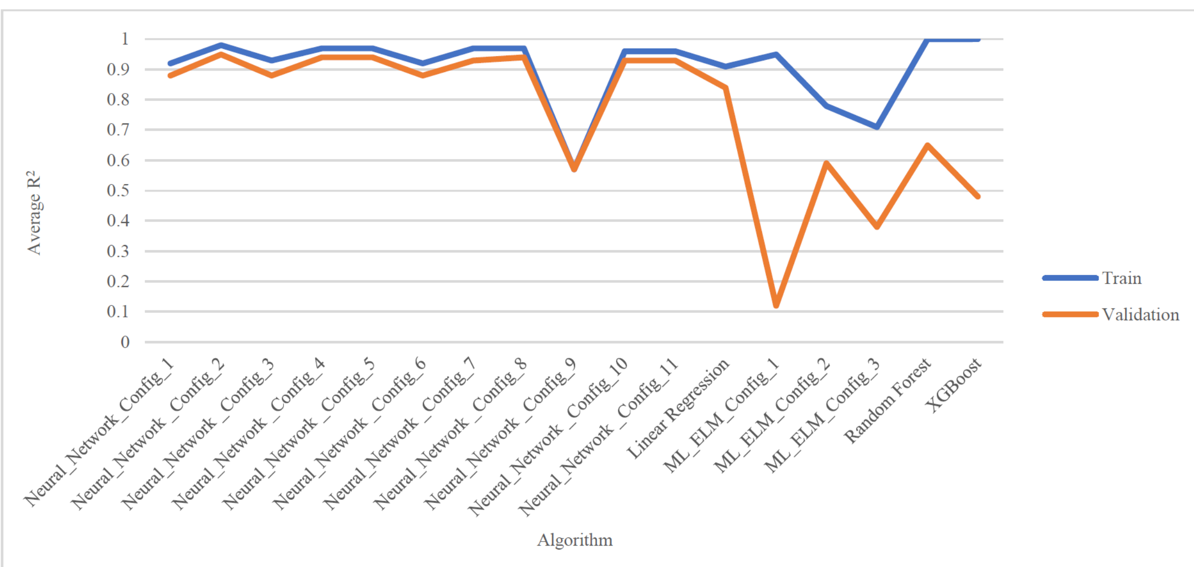

Figure 9—Comparison of average R^2 of training and validation datasets[9]

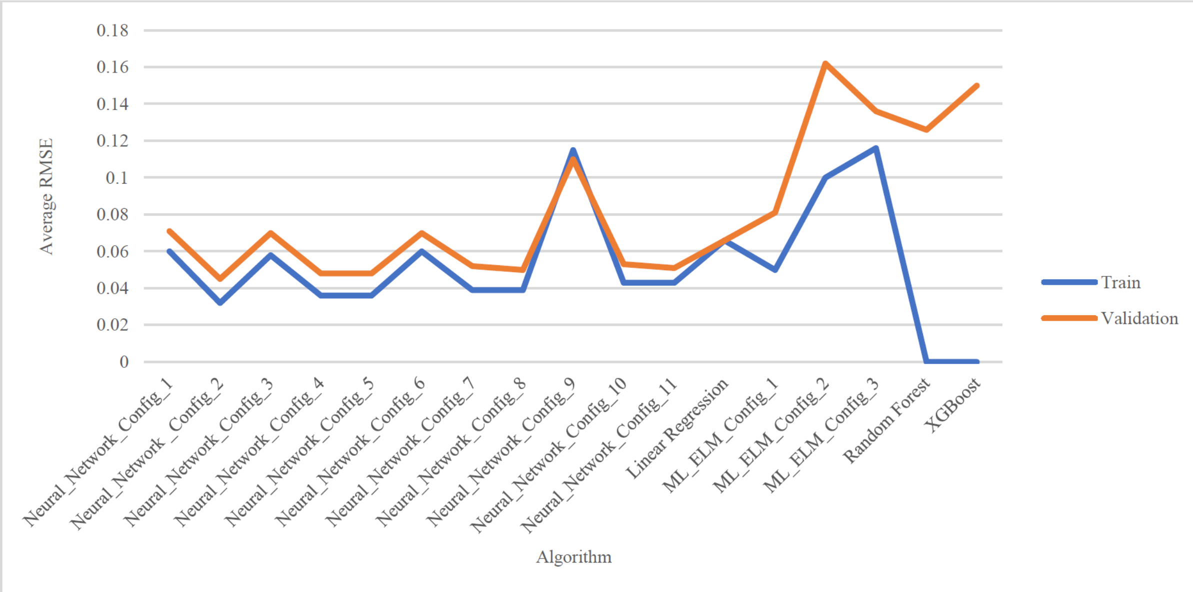

Figure 10—Comparison of average RMSE of training and validation datasets[9]

Neural Network configurations demonstrate remarkably stable performance with minimal gaps between training and validation RMSE and R^2 scores. Most of the neural networks maintain validation R^2 above 0.85-0.95, indicating these models successfully learned generalizable patterns rather than memorizing training data. Tree-based algorithms like Random Forest and XGBoost show the most dramatic performance drops - their training R^2 scores near 1.0 plummet to 0.65 and 0.48, respectively, and their training RMSE scores increase drastically from 0.0 to 0.13 and 0.15, respectively, on validation. This represents classic overfitting, where models memorize training patterns but fail to generalize. ML-ELM configurations also display larger training-validation gaps compared to neural networks. Linear Regression shows consistent performance between training and validation (both R^2 around 0.83-0.90). Thus Neural Networks are the clear winner; they consistently outperform all other models. They successfully learned generalizable patterns rather than memorizing training data. Tree-based models show severe overfitting. Multi-layer ELM shows inconsistent performance.

Out-of-sample validation

The performance of all trained models on the 15% hold-out test set is shown in Table 6 and Fig. 11.

Table 6—Model's performance on the unseen test data[9].

Model	R^2	MSE	MAE	RMSE
Neural_Network_Config_1	0.83	0.0099	0.075	0.099
Neural_Network_Config_2	0.93	0.0042	0.004	0.065
Neural_Network_Config_3	0.85	0.0088	0.070	0.094
Neural_Network_Config_4	0.92	0.0048	0.048	0.070
Neural_Network_Config_5	0.91	0.0054	0.049	0.073
Neural_Network_Config_6	0.83	0.0100	0.070	0.100
Neural_Network_Config_7	0.89	0.0062	0.064	0.078
Neural_Network_Config_8	0.92	0.0049	0.053	0.070
Neural_Network_Config_9	0.89	0.0066	0.051	0.081
Neural_Network_Config_10	0.90	0.0057	0.067	0.073

Model	R^2	MSE	MAE	RMSE
Neural_Network_Config_11	0.91	0.0054	0.057	0.131
Linear Regression	0.84	0.0092	0.075	0.078
ML_ELM_Config_1	0.71	0.0170	0.093	0.422
ML_ELM_Config_2	-2.03	0.1784	0.185	0.268
ML_ELM_Config_3	-0.22	0.0720	0.173	0.096
Random Forest	0.90	0.0061	0.054	0.078
XGBoost	0.83	0.0098	0..076	0.099

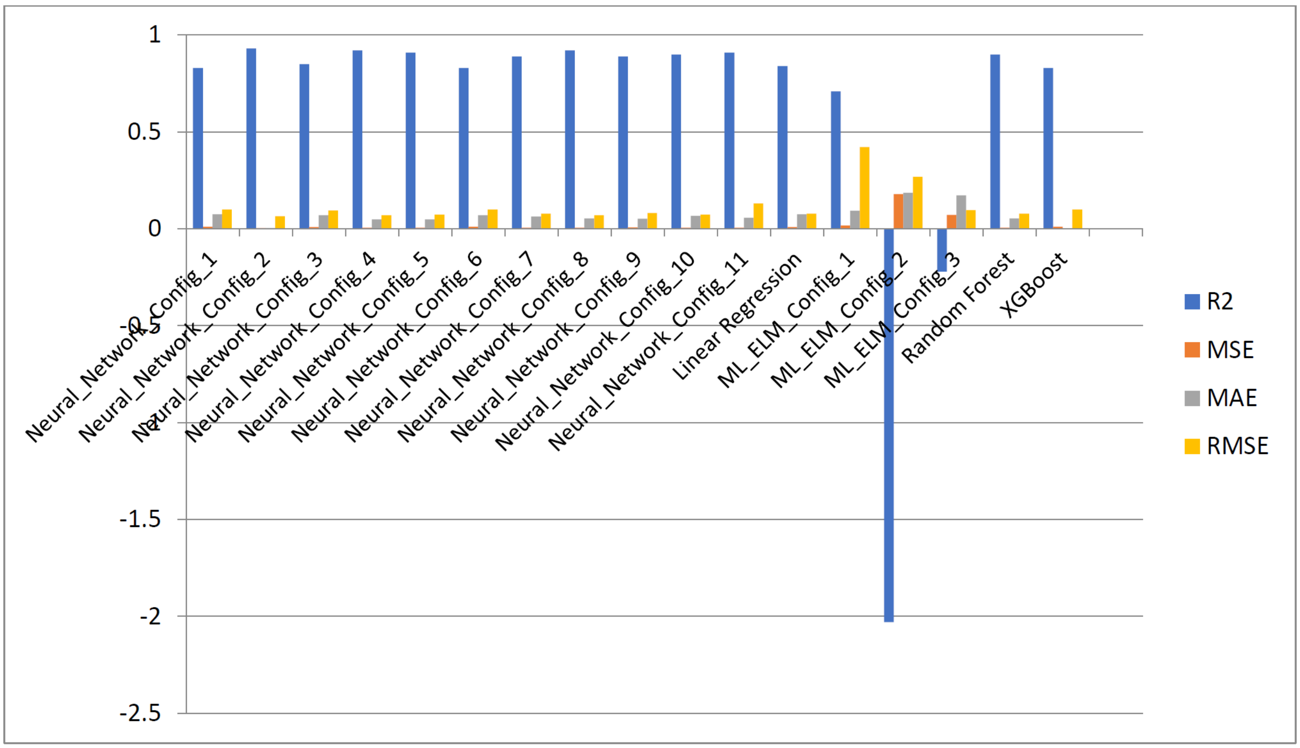

Figure 11—Model's performance on the unseen test data[9].

Since Neural Networks are able to predict the Geological Chance of Success better than rest models, therefore a comparative analysis is done and shown in Fig. 12. Config_2, 4, and 8 are the most reliable configurations as they have stable R^2 value across all the datasets. These different configurations are given in Table 7 and in Fig. 12.

Table 7—Comparative analysis of Configuration of three Neural Networks.

Configuration	Hidden Layers	Neurons Per layer	Hidden layer activation	Output layer activation	Optimization Algorithm	Learning rate	L2 regularization lambda(λ)
Neural_Network_Config_2	2	[32,16]	ReLU	Linear	Adam	0.001	0.01
Neural_Network_Config_4	3	[64,32,64]	ReLU	Linear	Adam	0.001	0.01
Neural_Network_Config_8	3	[64,32,16]	ReLU	Linear	Adam	0.001	0.01

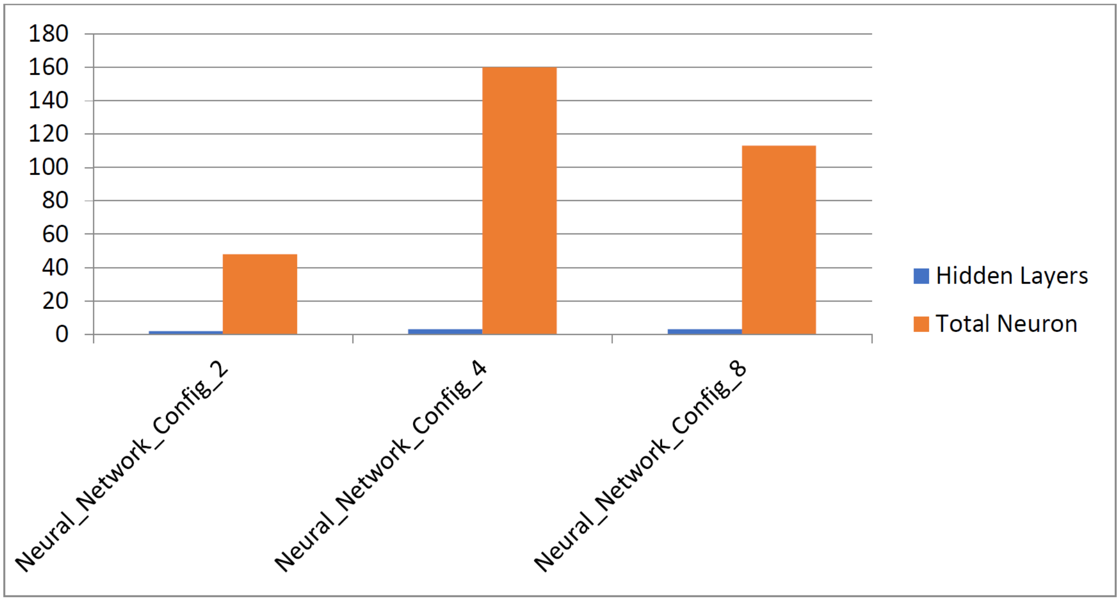

Figure 12—Representation of Configuration of three Neural Networks[9].

Table 8 presents the different R square for training, test and validation datasets for all the models tested in the study. Neural_Network_Config_2 emerging as the most optimal model due to its lowest variation in R^2 value, indicating its excellent generalization stability as shown in Fig.13.

Table 8—Comparative analysis of R^2 of different Neural Network Configuration

Model	Training R^2	Validation R^2	Test R^2
Neural_Network_Config_1	0.92	0.88	0.83
Neural_Network _Config_2	0.98	0.95	0.93
Neural_Network_Config_3	0.93	0.88	0.85
Neural_Network_Config_4	0.97	0.94	0.92
Neural_Network_Config_5	0.97	0.94	0.91
Neural_Network_Config_6	0.92	0.88	0.83
Neural_Network_Config_7	0.97	0.93	0.89
Neural_Network_Config_8	0.97	0.94	0.92
Neural_Network_Config_9	0.57	0.57	0.89
Neural_Network_Config_10	0.96	0.93	0.90
Neural_Network_Config_11	0.96	0.93	0.91
Linear Regression	0.91	0.84	0.84
ML_ELM_Config_1	0.95	0.12	0.71
ML_ELM_Config_2	0.78	0.59	-2.03
ML_ELM_Config_3	0.71	0.38	-0.22
Random Forest	1.00	0.65	0.90
XGBoost	1.00	0.48	0.83

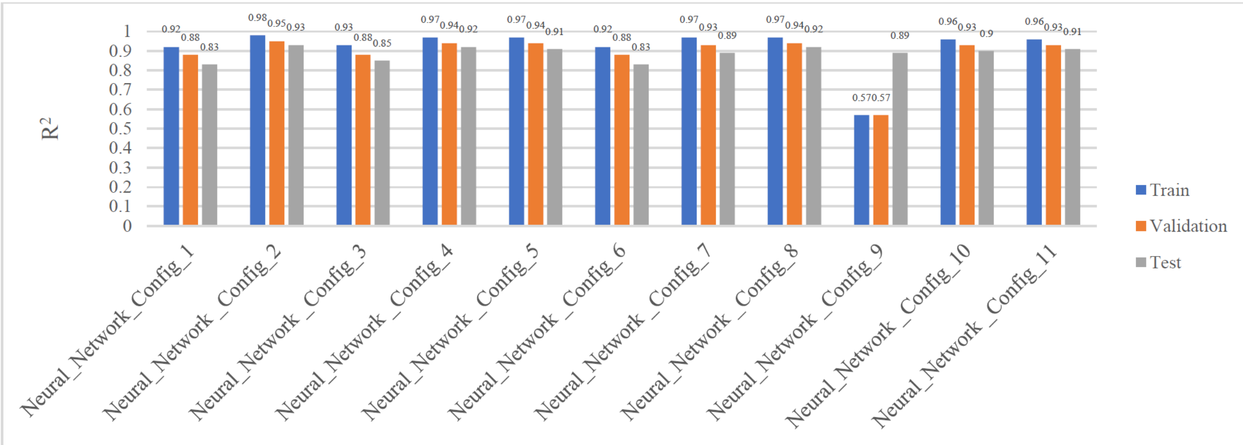

Figure 13—Comparative analysis of R^2 of different Neural Network Configuration[9]

Discussion

The implementation of AI and machine learning fundamentally changes the process of geological site screening for Carbon Capture, Utilization, and Storage (CCUS), giving us far-reaching impacts both for project developers and the wider society. By deploying advanced neural networks trained on large, well-curated geoscience datasets, the framework explained in this study enables rapid assessment of subsurface sites, drastically reducing the evaluation timeline from months or years to just weeks or even days. This acceleration allows oil and gas operators, investors and emitters to focus their technical and financial efforts only on the highest-potential candidates, thereby de-risking capital allocation and improving the overall economics of CCUS storage projects like Acorn project in the UK North Sea[7].

ML algorithms are computationally efficient because they do not impose the strong assumptions on the data-generating process that standard statistical or engineering procedures require[9]. The method uses cross-validation and focuses on geologically meaningful features, helping reduce human bias and subjectivity in site selection. As a result, operators can trust the predictive power of these AI models, which have demonstrated superior generalization to unseen data compared to tree-based algorithms that often overfit in complex geological contexts.

Moreover, the scalability and repeatability of this approach encourages stakeholders to evaluate vast geographic areas and build robust storage site assessments aligned with their decarbonization goals.

That said, challenges remain. Another limitation[7] is that geological chance of success will vary from field to field, with complex data dependencies that are often difficult to resolve without expert interpretation[8]. In many cases, datasets are not publicly available to validate key parameters such as containment capacity, and public geoscience databases remain scarce. Furthermore, training AI-driven models requires large, high-quality datasets, but for many oil and gas fields and other viable storage sites esp aquifers, the data either does not exist or is not published making it difficult to achieve consistent global applicability.

In conclusion, AI and machine learning have the potential to transform CCUS site screening by accelerating timelines, reducing bias, and improving economic outcomes. But for this promise to be fully realized, the industry must confront the challenges of data scarcity, regional variability, and the need for transparency. The entire CCUS storage study schedule and timing is indicative and is subject to the size of the area of interest being studied, the number of fields being evaluated, and the availability of legacy data (wells, production history) and geological/simulation models.

Conclusion

This study successfully developed and validated a data-driven AI/ML framework to accelerate the preliminary screening of geological CO_2 storage sites. The research demonstrated that deep neural networks are a superior and highly effective tool for this task, consistently achieving excellent performance ($R^2 >$ 0.90) across training, validation, and unseen test data. This indicates their ability to learn and generalize the complex, nonlinear relationships inherent in geological systems.

A critical finding of this work is the pronounced overfitting exhibited by popular tree-based algorithms like Random Forest and XGBoost. While these models achieved near-perfect scores on training data, their performance degraded significantly on validation and test sets, proving them unreliable for this application. This result highlights that robust cross-validation is not merely best practice but essential for identifying flawed models and ensuring that predictive tools are genuinely effective.

Ultimately, this AI-driven methodology provides the CCUS industry with a powerful tool to de-risk and accelerate the site selection process. By automating data integration, pattern recognition, and risk evaluation, this site selection methodology can reduce the preliminary screening timeline from several months to only a few days, thereby significantly lowering project costs. By enabling the rapid and objective ranking of basins based on geological merit, it allows operators to focus capital-intensive characterization efforts on the most promising candidates. This work establishes a foundational model that can be scaled with global datasets, paving the way for faster, more efficient, and more reliable deployment of carbon storage projects worldwide.

Acknowledgement

We thank all individuals who provided their expertise and support to this study. This research was made possible by the use of open-source data and models from various organisations. We also acknowledge the assistance of AI large language models for improving the clarity and readability of this manuscript.

References

1. Sarker, I.H. Deep Learning: A Comprehensive Overview on Techniques, Taxonomy, Applications and Research Directions. *SN COMPUT. SCI*. **2**, 420 (2021).
2. J. Tang, C. Deng and G. -B. Huang, "Extreme Learning Machine for Multilayer Perceptron," in *IEEE Transactions on Neural Networks and Learning Systems*, vol. **27**, no. 4, pp. 809–821, April 2016, doi: 10.1109/TNNLS.2015.2424995.
3. Breiman, L. Random Forests. *Machine Learning* **45**, 5–32 (2001).
4. Tianqi Chen, *University of Washington; Carlos Guestrin*, University of Washington. XGBoost: A Scalable Tree Boosting System, arXiv:1603.02754
5. Mustafa & Okur, Mehmet. (2022). A Comprehensive Review of Feature Selection and Feature Selection Stability in Machine Learning. *GAZI UNIVERSITY JOURNAL OF SCIENCE*. **36**. 10.35378/gujs.993763.
6. Schober, Patrick MD, PhD, MMedStat; Boer, Christa PhD, MSc; Schwarte, Lothar A. MD, PhD, MBA. Correlation Coefficients: Appropriate Use and Interpretation. *Anesthesia & Analgesia* **126**(5): p 1763–1768, May 2018. | DOI: 10.1213/ANE.0000000000002864
7. Juan Alcalde (2021). A criteria-driven approach to the CO2 storage site selection of East Mey for the Acorn project in the North Sea. *Marine and Petroleum Geology*
8. Bachu, S., 2000. Sequestration of CO2 in geological media: criteria and approach for site selection in response to climate change. *Energy Convers. Manag*. **41**, 953–970. https://doi.org/10.1016/S0196-8904(99)00149-1.
9. *Internal document*, Rara Energy Consulting, 2025
10. *Geoscience Australia, Data Discovery Portal*. https://portal.ga.gov.au